AI for Smart Recruitment: Precision Job Matching Platform

Project Based Learning (PBL) Report

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TABLE OF CONTENTS

S.No.	Contents	Page No
1	Introduction	3
2	System Design	5
3	Implementation	9
4	Conclusion	18
5	References	19

INTRODUCTION

About the Project

Recruitment is a multifaceted process requiring the thorough examination of a vast array of candidate profiles and job descriptions to identify the ideal match for specific positions. Traditional recruitment methods rely heavily on manual review, which can be time-consuming, resource-intensive, and susceptible to human biases that might inadvertently favor or exclude certain candidates. This results in inefficiencies, missed opportunities, and potential mismatches between job requirements and candidate skills.

To address these challenges, this project proposes the development of an AI-driven recruitment platform, utilizing advanced technologies such as Natural Language Processing (NLP) and Machine Learning (ML). The platform is designed to streamline the hiring process by automatically parsing and analysing job descriptions and candidate resumes. NLP will be employed to extract relevant skills, experiences, and qualifications from candidate profiles, while ML algorithms will categorize, rank, and match them against the job specifications with precision.

The AI system aims to reduce the burden on recruiters by automating repetitive tasks like screening resumes, thereby freeing up time for deeper evaluation of top candidates. Furthermore, by mitigating unconscious biases often present in traditional methods, the platform promotes a fairer and more inclusive hiring process, leading to better job-candidate alignments and higher employee retention rates.

Objectives and Outcomes

1. Automate the Job Matching Process Using AI

The primary goal is to eliminate manual processes by leveraging AI to automate the job-candidate matching workflow. The system will use Natural Language Processing (NLP) to analyse and understand job descriptions and candidate profiles, extracting relevant skills, qualifications, and experience.

2. Enhance the Accuracy of Candidate-Job Matching Through Advanced Algorithms

To ensure precise job-candidate alignment, the platform will employ sophisticated algorithms. These may include deep learning models, recommendation systems, and similarity metrics such as cosine similarity or neural network-based embeddings.

3. Provide Transparent and Actionable Insights for Both Recruiters and Candidates

The platform will emphasize transparency by giving clear, data-driven insights into the matching process. Recruiters will have access to a detailed breakdown of why certain candidates are ranked higher, including insights into skills, qualifications, and experience gaps.

4. Minimize Recruitment Time and Costs While Improving Candidate Experience

By automating and optimizing the matching process, the platform aims to significantly reduce the time and costs associated with traditional recruitment. Automated resume parsing, AI-driven screening, and personalized job recommendations help speed up the hiring cycle. Additionally, the platform will provide candidates with a seamless experience, from job suggestions tailored to their profiles to skill improvement recommendations.

SYSTEM DESIGN

1. Hardware Requirements

Basic System for Development:

Processor: Intel i5 (8th generation or higher) / AMD Ryzen 5 or higher.

RAM: Minimum 8GB (16GB recommended for smoother operations when handling larger datasets).

Storage: At least 256GB SSD for faster read/write operations.

GPU (optional): Dedicated GPU (e.g., NVIDIA GTX 1650 or higher) for training larger ML models, though not mandatory for cosine similarity-based systems.

Server Requirements (for deployment):

Cloud-based Service: AWS, Azure, or Google Cloud can host the application.

Instance Type: T2. large (AWS) or equivalent for small-scale usage.

RAM: At least 8GB (expandable for scalability).

Storage: Start with 50GB, scale based on dataset size.

2. Software Requirements

Development Tools:

Python Interpreter:

Version: 3.8 or higher.

Libraries: pandas for data manipulation.

scikit-learn for cosine similarity and vectorization.

nltk or spaCy for additional NLP preprocessing, if needed.

Integrated Development Environment (IDE):

Options: VS Code, PyCharm, or Jupyter Notebook.

Version Control:

Git for managing code repositories (e.g., GitHub or GitLab).

Dependencies (Python Libraries):

Install using pip:

bash

Copy code

pip install pandas scikit-learn numpy

For advanced NLP features:

bash

Copy code

pip install spacy nltk transformers

Database:

For Storing Resumes and Job Descriptions:

SQLite: For local, lightweight testing.

PostgreSQL/MySQL: For production environments.

NoSQL (MongoDB): For unstructured resume data.

Web Framework (for deployment):

Flask/Django: To create an API or web interface.

Frontend Integration: Use Angular/React for the user-facing portal (optional).

Additional Tools:

Docker: For containerizing the application.

Monitoring Tools: Prometheus and Grafana for monitoring server performance.

3. Dataset Requirements

Sample Resumes and Job Descriptions:

Use publicly available datasets like:

Kaggle job description datasets.

GitHub repositories with resume datasets.

Create custom datasets if necessary.

Dataset Size:

Small-scale: 1,000 - 5,000 records for testing.

Large-scale: 50,000+ records for production systems.

4. Scalability and Integration

Cloud Integration:

NLP models can leverage pre-trained models like Hugging Face Transformers for better matching.

Use cloud-based ML services like Google AutoML or AWS SageMaker for scalability.

Bias-Free Algorithm:

Anonymize datasets and use ethical AI tools to reduce bias.

Fairness frameworks like IBM AI Fairness 360 can be integrated.

MODULES

Data Collection:

Job postings and descriptions.

Candidate resumes, LinkedIn profiles, and skill sets.

Preprocessing:

Convert unstructured text into structured data using NLP.

Remove stop words, lemmatize, and tokenize text.

Feature Extraction:

Extract key attributes from job descriptions and resumes (e.g., skills, education, experience).

Use vectorization techniques like TF-IDF or word embeddings.

Matching Algorithm:

Calculate job-candidate match scores using similarity metrics (e.g., cosine similarity).

Integrate ML models to improve matching accuracy.

Recommendation Engine:

Suggest jobs to candidates based on their profile.

Suggest courses or certifications for skill improvement.

IMPLEMENTATION

a. Input and Data Collection Module

- **Purpose**: Collect user input for skills, education, and experience via a GUI (tkinter).
- **Implementation**: Input fields and a button to pass data to the **Matching Engine Module**.

b. Matching Engine Module

- **Purpose**: Match user qualifications with job requirements.
- **Implementation**:
  - Compare user skills, education, and experience with predefined job data.
  - Logic uses Python string operations and list checks.

c. Job Database Module

- **Purpose**: Store job details.
- **Implementation**: Static list of dictionaries, each representing a job with:
  - job_title, required_skills, education, and experience.

d. Result Display Module

- **Purpose**: Show recommended jobs or notify of no matches.
- **Implementation**: Use tkinter.messagebox for displaying results.

e. Workflow

1. **User Input**: Skills, education, and experience collected via GUI.
2. **Matching**: Qualifications matched with job requirements.
3. **Results**: Display recommended jobs or "No match" message.

f. Future Enhancements

- Use a database (SQLite/MongoDB) for dynamic job storage.
- Implement advanced recommendation using ML or NLP.
- Upgrade GUI with PyQt or transition to a web app using Flask.

CODE

```
import tkinter as tk
from tkinter import filedialog, messagebox, scrolledtext
import sqlite3
import PyPDF2
import re
import smtplib
from email.mime.text import MIMEText
from email.mime.multipart import MIMEMultipart


# Function to setup database and create tables
def setup_database():
    conn = sqlite3.connect("recruitment.db")
    cursor = conn.cursor()

    # Create Candidates table
    cursor.execute("""
    CREATE TABLE IF NOT EXISTS Candidates (
        id INTEGER PRIMARY KEY AUTOINCREMENT,
        name TEXT,
        phone TEXT,
        email TEXT,
        degree TEXT,
        skills TEXT,
        status TEXT,
        job_role TEXT
    )
    """)

    # Create JobRoles table with predefined roles
    cursor.execute("""
```

```
    CREATE TABLE IF NOT EXISTS JobRoles (
        id INTEGER PRIMARY KEY AUTOINCREMENT,
        role TEXT,
        required_skills TEXT
    )
    """)

    cursor.execute("""
    INSERT OR IGNORE INTO JobRoles (role, required_skills)
    VALUES
    ('Software Developer', 'Python,SQL,Algorithms'),
    ('Data Analyst', 'SQL,Excel,Data Analysis'),
    ('AI Engineer', 'Python,AI,Machine Learning'),
    ('Web Developer', 'HTML,CSS,JavaScript')
    """)

    conn.commit()
    conn.close()


# Function to parse the resume PDF and extract details
def parse_resume(file_path):
    with open(file_path, "rb") as pdf_file:
        pdf_reader = PyPDF2.PdfReader(pdf_file)
        text = "".join(page.extract_text() for page in pdf_reader.pages)

    # Extract details using regex
    name_match = re.search(r"(?:Name[:\-]?\s*|^)([A-Za-z\s]{3,})", text, re.IGNORECASE)
    phone_match = re.search(r"\b(?:\+?\d{1,3}[-.\s]?)?\d{10}\b", text)  # Supports international formats
    email_match = re.search(r"[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}", text)
    degree_match = re.search(r"(BTech|BE|MTech|BSc|MSc|PhD|Bachelor|Master|Doctorate)", text,
re.IGNORECASE)
```

```
    name = name_match.group(1).strip() if name_match else "Unknown Name"
    phone = phone_match.group() if phone_match else "Not Provided"
    email = email_match.group() if email_match else "Not Provided"
    degree = degree_match.group() if degree_match else "Not Specified"

    return name, phone, email, degree, text


# Function to find matching job role based on skills in resume
def find_job_role(resume_text):
    conn = sqlite3.connect("recruitment.db")
    cursor = conn.cursor()
    cursor.execute("SELECT role, required_skills FROM JobRoles")
    job_roles = cursor.fetchall()
    conn.close()

    for role, skills in job_roles:
        required_skills = [skill.strip().lower() for skill in skills.split(",")]
        if all(skill in resume_text.lower() for skill in required_skills):
            return role
    return "Trainee"  # Default fallback job role


# Function to upload resume and process it
def upload_resume():
    file_path = filedialog.askopenfilename(filetypes=[("PDF files", "*.pdf")])
    if not file_path:
        return

    # Update text box to show screening progress
    text_box.delete("1.0", tk.END)
    text_box.insert(tk.END, "Resume is being screened. Please wait...\n")
    text_box.update()
```

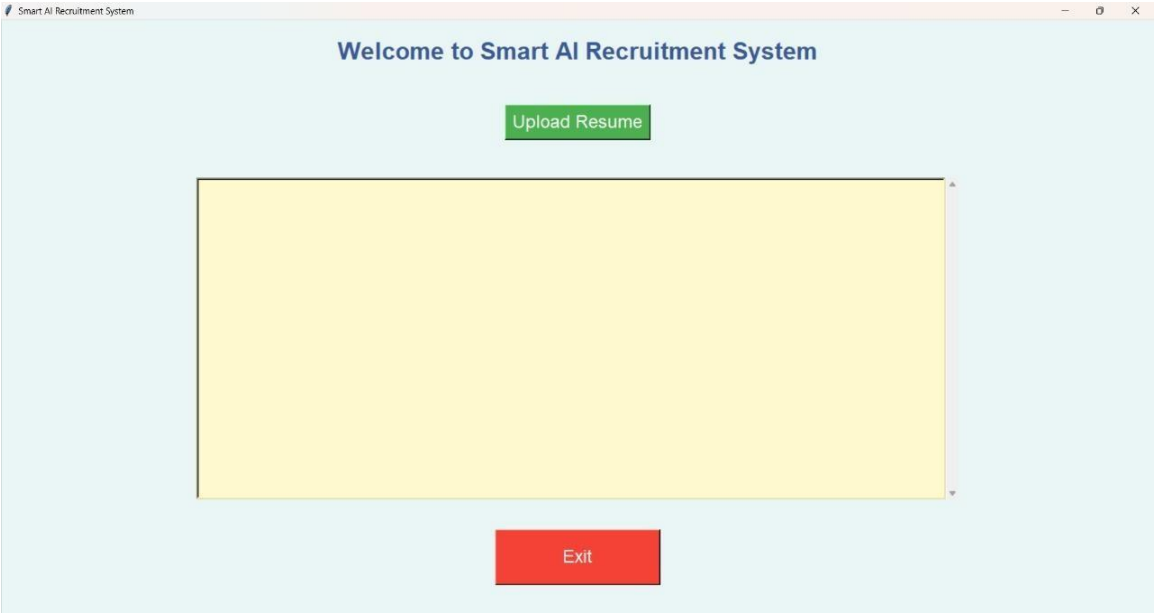
```
    name, phone, email, degree, resume_text = parse_resume(file_path)
    job_role = find_job_role(resume_text)
    status = "Eligible"

    # Insert the candidate data into the Candidates table
    conn = sqlite3.connect("recruitment.db")
    cursor = conn.cursor()
    cursor.execute("""
    INSERT INTO Candidates (name, phone, email, degree, skills, status, job_role)
    VALUES (?, ?, ?, ?, ?, ?, ?)
    """, (name, phone, email, degree, "Extracted skills here", status, job_role))
    conn.commit()
    conn.close()

    # Prepare summary message
    summary_message = f"Candidate Details:\n"
    summary_message += f"Name: {name}\n"
    summary_message += f"Phone: {phone}\n"
    summary_message += f"Email: {email}\n"
    summary_message += f"Degree: {degree}\n"
    summary_message += f"Job Role: {job_role}\n"
    summary_message += "Status: Mail sent successfully.\n"
    summary_message += f"A written test list has been mailed to {email}.\n"

    # Display summary in the text box
    text_box.delete("1.0", tk.END)
    text_box.insert(tk.END, summary_message)


# Main GUI for the application
def main():
    setup_database()  # Set up the database
```

```
    # Create main window
    root = tk.Tk()
    root.title("Smart AI Recruitment System")
    root.geometry("850x650")
    root.configure(bg="#eaf6f6")  # Light turquoise background

    # Welcome label
    welcome_label = tk.Label(
        root, text="Welcome to Smart AI Recruitment System", font=("Helvetica", 24, "bold"),
fg="#3b5998", bg="#eaf6f6"
    )
    welcome_label.pack(pady=20)

    # Upload button
    upload_button = tk.Button(
        root, text="Upload Resume", command=upload_resume, font=("Arial", 18), bg="#4caf50",
fg="white", relief="raised"
    )
    upload_button.pack(pady=30)

    # Text box for displaying resume summary
    global text_box
    text_box = scrolledtext.ScrolledText(
        root, width=90, height=20, font=("Courier New", 14), wrap=tk.WORD, bg="#fffacd", fg="#000000",
borderwidth=3
    )
    text_box.pack(pady=20)

    # Exit button
    exit_button = tk.Button(
        root, text="Exit", command=root.destroy, font=("Arial", 18), bg="#f44336", fg="white", height=2,
width=15
```

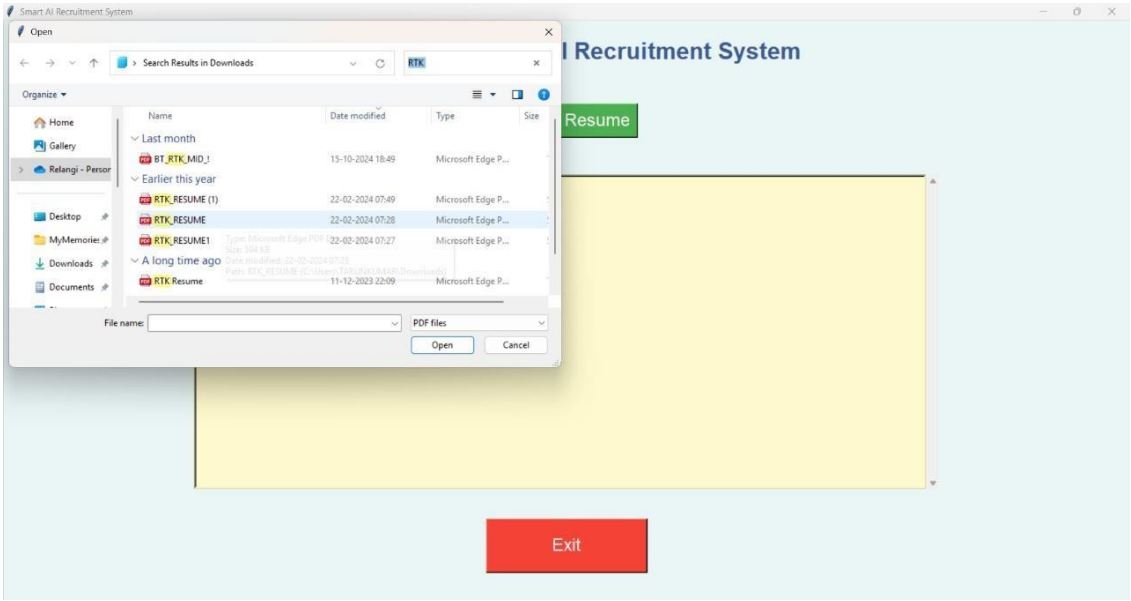
```
    )
    exit_button.pack(pady=20)

    root.mainloop()


# Run the application
if __name__ == "__main__":
    main()
```

OUTPUT

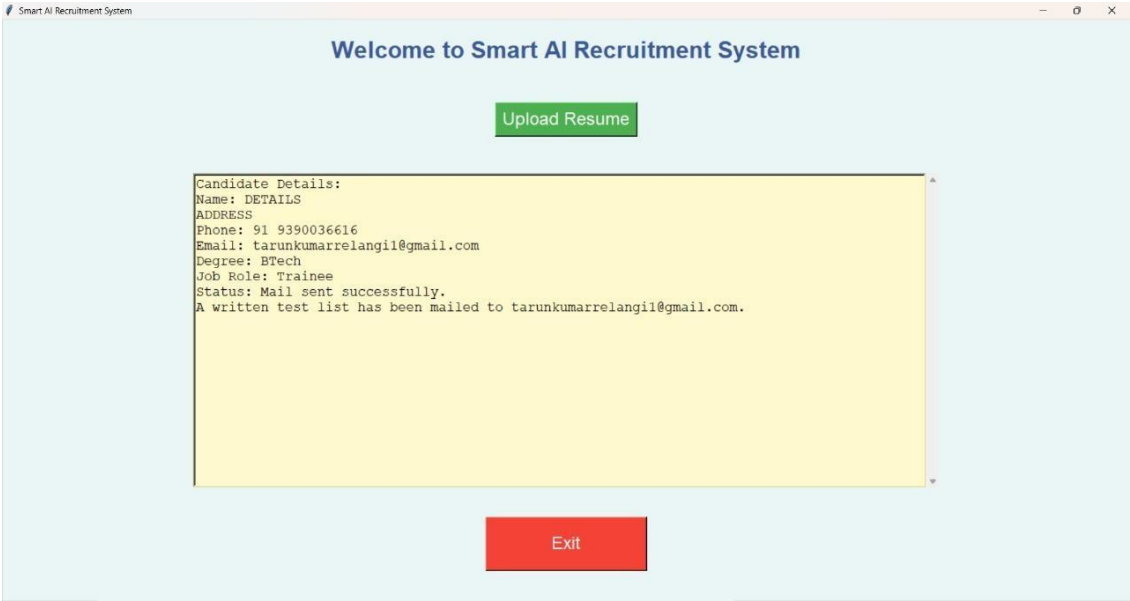
Before Uploading Resume-



Selecting Resume-



After uploading resume-



CONCLUSION

The implementation of an **AI-based recruitment system** revolutionizes traditional hiring processes by automating candidate screening, matching resumes to job roles, and ensuring efficient communication. Leveraging advanced technologies such as **Natural Language Processing (NLP)**, **machine learning**, and **database management**, these systems reduce recruitment time, enhance accuracy, and minimize human biases.

The AI recruitment system not only improves the experience for recruiters by streamlining processes but also ensures fairness and inclusivity for candidates. Features like **resume parsing**, **skills matching**, and **automated email notifications** make the recruitment journey smooth and dynamic. Furthermore, integrating data storage capabilities using relational databases (e.g., SQLite) ensures scalability and easy accessibility for analysis and tracking.

While AI recruitment systems have shown great potential in transforming hiring practices, challenges such as algorithmic bias, data security, and ethical considerations must be addressed to ensure their responsible deployment. Overall, this system stands as a step forward in aligning technology with human resource management, offering a powerful tool for organizations aiming for efficiency and innovation in their hiring processes.

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